F3: Fourier spectrum and additive energy of coordinate functions along $\langle G \rangle$ on prime-order curves over F_p

Run F3-RUN-02 of study `coord_spectrum_20261110`. Class: boundary diagnostic; exact counts, no S, no rho row, no leaderboard change. **Status: closed by decision rule 3; no replication is prescribed.** All numbers below are `[pcc:unverified]` (no pcc claim id exists for this run). No improvement over rho, or over anything, is claimed.

Purpose and mechanism

For a cheap $g : \langle G \rangle \to F_p$ the study measures $S_{g(a,b)} = \sum_{k \in \mathbb{Z}/r} e(ag([k]G)/p + bk/r)$ and $\eta_g = \max|S_{g(a,b)}| \ / \ r$. A coefficient $\eta \gg r^{-1/4}$ would give the logarithm by a hidden-shift step in about $\log r \ / \ \eta^2$ group operations (generator's derivation, derived, not checked). The bound it must escape: for g of algebraic degree d in the coordinates the Weil bound gives $\eta \lesssim 2dr^{-1/2}$ (Kohel–Shparlinski, recalled), which restates rho's barrier; only functions of degree about $r^{1/4}$ or more that stay cheap could escape it.

The second statistic is the additive energy $E(A) = \#\{(a,b,c,d) \in A^4 : a + b = c + d \text{ in } F_p\}$ of $A_K = \{x([k]G) : 1 \le k \le K\}$, which single character sums do not control: an excess over random growing with p would mean Freiman-type structure in the orbit (4-term x-coordinate relations harvestable without summation polynomials). Instrument: native Rust, exact-arithmetic (f64) FFT of length r at 64 values of a; exact energies by hashing pair sums for $K \in \{\lceil r^{1/2} \rceil, \lceil r^{2/3} \rceil\}$; nulls: 20 per class (spectrum), 50 random K-subsets of F_p (energy).

Rows and functions

R16, R20: 8 prime-order curves each, p the largest prime below 2^{16} / 2^{20}. **ANOM:** 2 anomalous curves ($\#E = p$), CM with $D = -11$, $p = 74869$ and $p = 1060513$ (the "2^{16}" curve lands above 2^{16}); positive control. Functions per curve: x, y, $\chi(x + c)$ for $c = 0, 1, 2$ (Legendre), Dickson $D_{n(x)}$ of degree $n = \lceil r^{1/4} \rceil$ (the Weil-escape class). Controls: λ the Smart p-adic logarithm (ANOM only; must be linear in k) and PLANT, $g(k) = \lfloor kp \ / \ r \rfloor$ (every curve).

Frozen predictions and rules (PROTOCOL.md sections 3 and 4, verbatim)

- **P1.** $\eta_x, \eta_y, \eta_\chi, \eta_D$ within $[0.8, 1.25] \times \text{median}(\eta_{\text{NULL}})$ on every R16/R20 curve. **P2.** λ on ANOM and PLANT everywhere: $\eta \ge 0.5$. **P3.** $E(A_K) \ / \ \text{mean} \ E(\text{random})$ within $1 \pm 3\frac{\text{sd}}{\text{mean}}$ of the null on every curve and K. **P4.** ANOM's x, y, χ rows behave as P1 (anomaly lives in λ, not in coordinates). Prior that P1 or P3 fails on ≥ 2 curves of one size: 0.03.

1. **Instrument failure:** P2 fails, or FFT error bound exceeds $10^{-6}r$. Fix, re-freeze, re-run.
2. **Lead alive:** a non-control g has $\eta_g > 1.5 \times \text{median}(\eta_{\text{NULL}})$ on ≥ 2 curves of one size, or P3 fails in the same direction on ≥ 2 curves of one size. Then replicate on 16 fresh curves and at 2^{24} before any claim.
3. **Closed:** otherwise: no heavy Fourier coefficient and no additive-energy excess in the coordinate orbit for these function classes at $r \le 2^{20}$. Rule 3 does not bound functions outside the classes listed, higher Gowers norms (U^3 and up), or larger p.

Amendments before the run (section 6, summarised)

Predictions and rules were committed (`c8bbc80e1a`) before the tool existed. After the implementer's `--quick` (not committed, not scored), three corrections to the statistic and null, not to predictions or rules: (1) **$b = 0$ is excluded from the gated η**: on a prime-order curve $\langle G \rangle = E(F_p)$, so $S_{g(a,0)}$ is a character sum over the whole group and carries no logarithm information; `eta_b0` is reported separately. (2) **Legendre rows enter as complex ± 1 values** against a null of random ± 1 sequences (the map $\pm 1 \mapsto \{1, p-1\}$ was an encoding artefact). (3) **Nulls carry the negation symmetry**: x, χ, D_n even, y, λ odd; nulls per class (`even_fp`, `even_pm1`, `odd_fp`, 20 each); P1/P4 ratios are against the null of the function's class; PLANT keeps the uniform null.

Freeze note (before the run). The quick run read x 0.74–0.79, y 0.73, χ 0.455–0.464, D_n 0.99–1.04 of class-null medians, so P1′s band was "expected to **fail downward** on x, y, χ; it is kept as written and will be scored as failed." Reading to be tested, not a rule: a Weil-type bound $|S_{g(a,b)}| \le C\sqrt{p}$ for every b sits below the random extreme $\approx \sqrt{r \ln r}$, so orbit spectra are flatter than random; rule 2 fires only upward.

Results `[pcc:unverified]`

Source: `results/run.json`, run F3-RUN-02, 15,500.9 s on 4 threads, shared local host. η columns are `eta_over_null_median` (coordinate function over the median of its symmetry-class null, 20 nulls); $\lambda\eta$ and PLANT η are raw; energy columns are `ratio_to_null_mean` and z for $K_1 = \lceil r^{1/2} \rceil$, $K_2 = \lceil r^{2/3} \rceil$. P3: ratio within $1 \pm 3\frac{\text{sd}}{\text{mean}}$, equivalent to $|z| \le 3$; true in all 36 pairs.

R16 (8 curves, $p = 65521$)

curve	r	x	y	χ(x)	χ(x+1)	χ(x+2)	D_n	PLANT η	E ratio, z (K1)	E ratio, z (K2)
R16_0	65257	0.758	0.732	0.474	0.474	0.473	1.060	1.000000	K=256: 0.99791, −0.59	K=1621: 0.99990, −0.48
R16_1	65587	0.763	0.728	0.464	0.464	0.464	0.964	1.000000	K=257: 0.99654, −0.91	K=1627: 0.99937, −2.94
R16_2	65479	0.767	0.721	0.484	0.483	0.484	1.039	1.000000	K=256: 0.99575, −1.49	K=1625: 1.00034, +1.21
R16_3	65731	0.748	0.739	0.471	0.472	0.472	0.998	1.000000	K=257: 1.00722, +2.16	K=1629: 0.99970, −1.13
R16_4	65353	0.763	0.714	0.472	0.472	0.473	0.976	1.000000	K=256: 1.00010, +0.03	K=1623: 1.00006, +0.31
R16_5	65447	0.754	0.728	0.460	0.460	0.460	0.997	1.000000	K=256: 1.00393, +0.96	K=1625: 1.00016, +0.50
R16_6	65731	0.758	0.728	0.463	0.463	0.463	1.005	1.000000	K=257: 1.00158, +0.48	K=1629: 1.00020, +0.79
R16_7	65407	0.764	0.721	0.475	0.475	0.476	1.014	1.000000	K=256: 0.99476, −1.33	K=1624: 1.00029, +1.18

R20 (8 curves, $p = 1048573$)

curve	r	x	y	χ(x)	χ(x+1)	χ(x+2)	D_n	PLANT η	E ratio, z (K1)	E ratio, z (K2)
R20_0	1046897	0.708	0.682	0.415	0.415	0.415	0.999	1.000000	K=1024: 1.00075, +0.81	K=10311: 1.00000, +0.10
R20_1	1049437	0.709	0.683	0.416	0.416	0.416	0.972	1.000000	K=1025: 1.00032, +0.35	K=10327: 0.99998, −1.02
R20_2	1048633	0.704	0.677	0.409	0.409	0.409	1.105	1.000000	K=1025: 1.00040, +0.51	K=10322: 0.99999, −0.22
R20_3	1046711	0.711	0.688	0.408	0.408	0.408	0.964	1.000000	K=1024: 0.99804, −1.62	K=10310: 1.00006, +2.21
R20_4	1047883	0.711	0.679	0.411	0.411	0.411	1.010	1.000000	K=1024: 0.99940, −0.60	K=10317: 1.00002, +0.88
R20_5	1049173	0.702	0.676	0.404	0.404	0.404	1.002	1.000000	K=1025: 1.00030, +0.35	K=10326: 0.99997, −1.07
R20_6	1050391	0.698	0.692	0.420	0.420	0.420	1.005	1.000000	K=1025: 1.00038, +0.41	K=10334: 1.00002, +0.70
R20_7	1049387	0.704	0.676	0.419	0.419	0.419	1.016	1.000000	K=1025: 1.00101, +1.06	K=10327: 1.00003, +1.23

ANOM (2 anomalous curves)

curve	r	x	y	χ(x)	χ(x+1)	χ(x+2)	D_n	λ η	PLANT η	E ratio, z (K1)	E ratio, z (K2)
ANOM_16	74869	0.774	0.733	0.474	0.474	0.474	0.991	1.000000	1.000000	K=274: 1.00468, +1.26	K=1777: 0.99998, −0.09
ANOM_20	1060513	0.700	0.686	0.411	0.411	0.411	1.000	1.000000	1.000000	K=1030: 1.00105, +1.07	K=10400: 1.00001, +0.24

The three χ columns agree to within 0.001 on every curve, so their markers nearly coincide in the figure.

Summary block (verbatim, `results/run-02.stdout.txt`, "numbers, not verdicts")


```
summary (numbers, not verdicts):
  P1_coordinate_functions_R16: {   "curves": 8,   "pairs": 48,   "eta_over_null_median_min":
4.5968749246628116e-1,   "eta_over_null_median_max": 1.060196599948265e0,   "pairs_outside_0.8_1.25":
40,   "pairs_above_1.5": 0,   "curves_with_a_pair_above_1.5": 0  }
  P3_energy_R16: {   "curve_K_pairs": 16,   "outside_1_pm_3sd_over_mean": 0,   "energy_ratio_min":
9.947561178965965e-1,   "energy_ratio_max": 1.0072178245006844e0,   "max_abs_z": 2.942544493207049e0  }
  P1_coordinate_functions_R20: {   "curves": 8,   "pairs": 48,   "eta_over_null_median_min":
4.040048343673199e-1,   "eta_over_null_median_max": 1.1048936213795235e0,   "pairs_outside_0.8_1.25":
40,   "pairs_above_1.5": 0,   "curves_with_a_pair_above_1.5": 0  }
  P3_energy_R20: {   "curve_K_pairs": 16,   "outside_1_pm_3sd_over_mean": 0,   "energy_ratio_min":
9.980426291680091e-1,   "energy_ratio_max": 1.0010057999069402e0,   "max_abs_z": 2.2143126599736997e0  }
  P4_anom_coordinates_ANOM: {   "curves": 2,   "pairs": 10,   "eta_over_null_median_min":
4.1128832744454824e-1,   "eta_over_null_median_max": 7.744092055274636e-1,   "pairs_outside_0.8_1.25":
10,   "pairs_above_1.5": 0,   "curves_with_a_pair_above_1.5": 0  }
  P3_energy_ANOM: {   "curve_K_pairs": 4,   "outside_1_pm_3sd_over_mean": 0,   "energy_ratio_min":
9.999805518799264e-1,   "energy_ratio_max": 1.0046844423355137e0,   "max_abs_z": 1.2579179925218338e0  }
  P2_controls: {   "lambda_eta_min": 1.0000000000000002e0,   "lambda_curves": 2,   "plant_eta_min":
9.999999996168333e-1,   "plant_curves": 18,   "lambda_linear_violations_total": 0  }
  instrument: {   "parseval_max_relative_deviation_from_r2": 8.682964995996564e-16,
"fft_vs_naive_max_abs_error_over_r": 1.347539132337562e-15,   "fft_error_bound_1e-6_r_holds": true,
"rule1_instrument_failure_flag": false  }
```

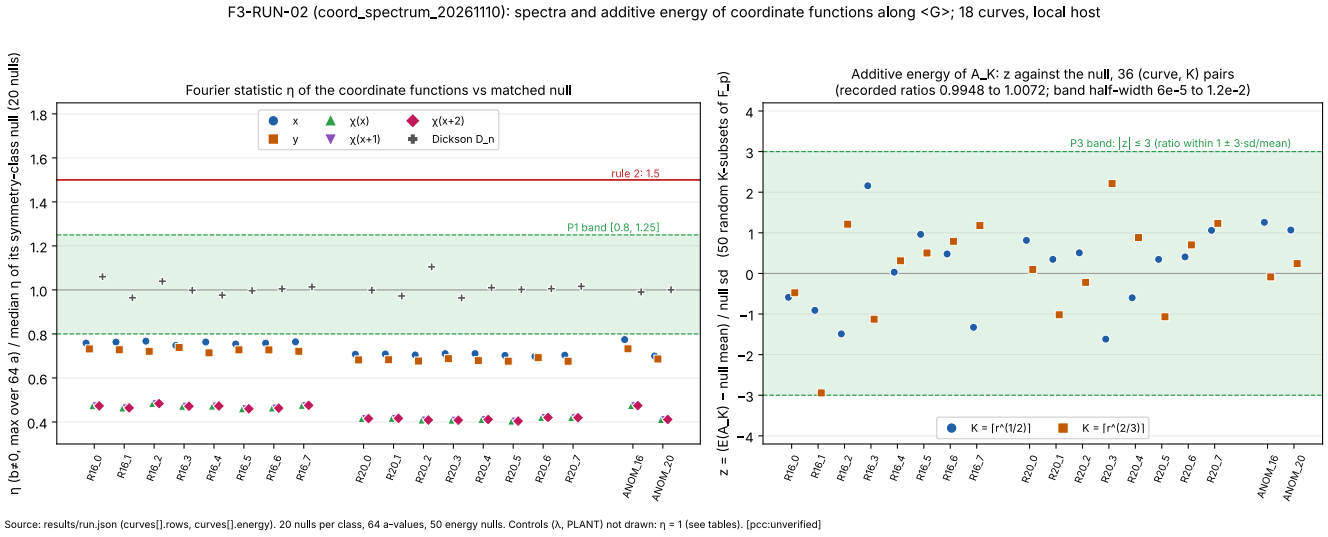



Figure 1: Left: η / median η of the matched symmetry-class null for $x, y, \chi(x), \chi(x+1), \chi(x+2), D_n$ on each curve; green band: P1 $[0.8, 1.25]$; red line: rule 2 at 1.5. Right: energy z per (curve, K) with the ± 3 band (equivalent to $1 \pm 3\frac{\mathrm{sd}}{\mathrm{mean}}$; recorded ratios 0.9948 to 1.0072). Run F3-RUN-02; 20 nulls per class, 64 values of a, 50 energy nulls; local shared host. Source: `results/run.json`; `figures/plot.py`.

Rule application (verbatim, session log 2026-11-12 00:05Z) [pcc:unverified]

Rules applied verbatim (§4, first applicable): rule 1 no (P2 holds, FFT bound holds); rule 2 no (no non-control g with $\eta > 1.5 \times$ null median on ≥ 2 curves of one size; P3 fails nowhere); **rule 3: closed** — "no heavy Fourier coefficient and no additive-energy excess in the coordinate orbit for these function classes at r $\leq 2^{20}$". P4 reads as P1 (ANOM coordinates flat; the anomaly lives in λ, $\eta = 1$). P1 as written fails: 40 of 48 pairs outside $[0.8, 1.25]$ on R16 and on R20, all below, as announced in the freeze note; on ANOM 10 of 10 below. P2 holds; P3 holds (0 of 36 outside).

Reading (offered, not a result, not a rule) [pcc:unverified]

The x, y, χ rows sit at 0.40–0.78 of their class-null medians: orbit spectra of these functions are flatter than random, consistent with a Weil-type bound $|S| \leq C\sqrt{p}$ for every b, which lies below the random extreme $\approx \sqrt{r \ln r}$. The Dickson rows (degree 16 to 33) sit at 0.96–1.10, at the null: a reading is that the degree-dependent constant exceeds the random maximum, leaving no room to flatten them. Not tested by a separate experiment.

Rule 3′s own caveats. The closure does not bound functions outside the listed classes, higher Gowers norms (U^3 and up), or larger p; $r \leq 2^{20}$ only; one Dickson degree per curve.

Limitations

Local host (4 vCPU, shared with three other running studies); the isolab-eks confirmatory run is open (no aws CLI in the session); counts exact, no timing result. 20 nulls per class, 64 values of a, 50 energy subsets. 8 + 8 + 2 curves; $c \in \{0, 1, 2\}$ for χ. R16_1 at K_2 has $z = -2.94$, 0.06 inside the band (all others $|z| \leq 2.22$); noted, not scored. No research-kb query; no novelty claim.

Evidence pointers

Card	`CARD-coord-spectrum-20261110-F3-RUN-02`, sha256 `d09186447235f7da48e4e1e0d188c90141206dff54661cec1e733dc054c66f3a`
Receipt	`cards/receipts/F3-RUN-02.receipt.json` (completed, 13 commands, each one attempt, exit 0)
Pinned head	`9465664dbe6ba5757b9cb011192215a00605e8d8`
Binary	`tool/target/release/coord_spectrum` sha256 `7e08c12ec3865247299923e9f71398260dfe605b261ce66c5a78226496fe1c4e`
run.json	sha256 `642bf9cd48c7f11fabc73f29c7563e488c14742148c5aa4f948ea52daa8dc30a` (151,455 bytes)
FREEZE.sha256	sha256 `4891d5aa563392fe32dd9c3035c2d65c43bd749af8efca329853facb8775bf7a`; PROTOCOL.md `4bfd872a04d0e7b671446c6016a541f2e7384fdecc411e1c4d05a0810be2ec7f`; tool/src/main.rs `ecb877ef619ea88e80fbe096059cd43bfb4022f67ecc1e5b2f3fc73c208289a6`; tool/Cargo.toml `a3f4cf8b04cd89c2b071ba0c27f113cd7f5c54559d064dc5fda807239c76f52a`; tool/Cargo.lock `3cc4284b70d5c7155750195fc11e5a1c8b66d4fcb25cd9886682c0adc3cfd203` (all OK in the receipt)
Tests	7 passed, 0 failed (`F3-RUN-02.test.stdout`)
Instrument	Parseval max rel. deviation 8.68e-16; FFT vs naive max $\lvert\Delta S\rvert/r$ 1.35e-15 (bound 10^{-6} holds); λ linearity violations 0
Report build	typst Python wheel 0.15.2 (scratch venv); figure by matplotlib 3.11.2